Using knit_expand() for templates

Yihui Xie

October 19, 2026

Contents

1 Write one row of data **1**
1.1 Now i is 1 . . . 1
1.2 Now i is 2 . . . 1
1.3 Now i is 3 . . . 1
1.4 Now i is 4 . . . 2
1.5 Now i is 5 . . . 2

2 A regression model on several variables **2**
2.1 Regression on cyl . . . 2
2.2 Regression on disp . . . 2
2.3 Regression on hp . . . 3
2.4 Regression on drat . . . 3
2.5 Regression on wt . . . 3
2.6 Regression on qsec . . . 4
2.7 Regression on vs . . . 4
2.8 Regression on am . . . 4
2.9 Regression on gear . . . 5
2.10 Regression on carb . . . 5

3 Multiple variables **5**

1 Write one row of data

Only the first two sections are evaluated.

1.1 Now i is 1

This chunk is evaluated.

```
# row number 1
iris[1, ]
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
```

1.2 Now i is 2

This chunk is evaluated.

```
# row number 2
iris[2, ]
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 2          4.9           3          1.4         0.2  setosa
```

1.3 Now i is 3

This chunk is not evaluated.

```
# row number 3
iris[3, ]
```

1.4 Now i is 4

This chunk is not evaluated.

```
# row number 4
iris[4, ]
```

1.5 Now i is 5

This chunk is not evaluated.

```
# row number 5
iris[5, ]
```

2 A regression model on several variables

You can expand a template file, or just provide the template as a character string.

```
# the template
tpl = c("\\subsection{Regression on {{xvar}}}", "<<lm-{{xvar}}>>=",
    "lm(mpg~{{xvar}}, data=mtcars)", "@")
# expand to knitr source and pass to knit()
src = lapply(names(mtcars)[-1], function(xvar) {
    knit_expand(text = tpl)
})
```

2.1 Regression on cyl

```
lm(mpg ~ cyl, data = mtcars)

##
## Call:
## lm(formula = mpg ~ cyl, data = mtcars)
##
## Coefficients:
## (Intercept)          cyl
##       37.88        -2.88
```

2.2 Regression on disp

```
lm(mpg ~ disp, data = mtcars)

##
## Call:
## lm(formula = mpg ~ disp, data = mtcars)
##
## Coefficients:
## (Intercept)         disp
##     29.5999      -0.0412
```

2.3 Regression on hp

```
lm(mpg ~ hp, data = mtcars)

##
## Call:
## lm(formula = mpg ~ hp, data = mtcars)
##
```

```
## Coefficients:
## (Intercept)           hp
##     30.0989      -0.0682
```

2.4 Regression on drat

```
lm(mpg ~ drat, data = mtcars)

##
## Call:
## lm(formula = mpg ~ drat, data = mtcars)
##
## Coefficients:
## (Intercept)         drat
##       -7.52         7.68
```

2.5 Regression on wt

```
lm(mpg ~ wt, data = mtcars)

##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Coefficients:
## (Intercept)           wt
##       37.29        -5.34
```

2.6 Regression on qsec

```
lm(mpg ~ qsec, data = mtcars)

##
## Call:
## lm(formula = mpg ~ qsec, data = mtcars)
##
## Coefficients:
## (Intercept)         qsec
##       -5.11         1.41
```

2.7 Regression on vs

```
lm(mpg ~ vs, data = mtcars)

##
## Call:
## lm(formula = mpg ~ vs, data = mtcars)
##
## Coefficients:
## (Intercept)           vs
##       16.62         7.94
```

2.8 Regression on am

```
lm(mpg ~ am, data = mtcars)

##
## Call:
## lm(formula = mpg ~ am, data = mtcars)
##
## Coefficients:
## (Intercept)           am
##       17.15         7.24
```

2.9 Regression on gear

```
lm(mpg ~ gear, data = mtcars)

##
## Call:
## lm(formula = mpg ~ gear, data = mtcars)
##
## Coefficients:
## (Intercept)         gear
##        5.62         3.92
```

2.10 Regression on carb

```
lm(mpg ~ carb, data = mtcars)
```

```
##
## Call:
## lm(formula = mpg ~ carb, data = mtcars)
##
## Coefficients:
## (Intercept)         carb
##       25.87        -2.06
```

3 Multiple variables

```
tpl = "The value of a is {{a}} and b is {{b}}."
```

The value of a is 1 and b is 2.
The value of a is 5 and b is 2013.